

Number theory

• Divisibility:

$$\forall a, b \in \mathbb{Z}$$

With $a \neq 0$, we say that a divides b (and we write $a|b$)

If $\exists$ (there exist) some $c \in \mathbb{Z}$ s.t

$$b = a \cdot c$$

Example

$$4|8 \quad (\because 8 = 4 \times 2) \quad \because 2 \in \mathbb{Z}$$

$$1|n \quad (\because 1 \times n = n)$$

$$\because a \nmid b \quad (a \text{ not divides } b)$$

Theorem:

Let $a, b, c \in \mathbb{Z}$ then

- (1) $a|0$ (2) $1|a$ and $-1|a$
 - (3) $a|a$ (4) If $a|b$ and $b|a$ then $a = b$
 - (5) If $a|b$ and $b|c$ then $a|c$
 - (6) If $a|b$ and $a|bc$
 - (7) If $a|b$ and $a|bc$
 - (8) If $a|b$ and $a|c$, then $a|bx + cy$
- for all $x, y \in \mathbb{Z}$

Proof:

Given that

$$a/0$$

we can write $0 = a \cdot 0$ for

some $0 \in \mathbb{Z}$

$$\Rightarrow a/0$$

$$1/a \text{ and } -1/a$$

We can write

$$a = 1 \cdot 0 \text{ and } a = -1 \times (-a)$$

$$\Rightarrow 1/a \text{ and } -1/0$$

$$a/0$$

We can write

$$a = a \times 1 \text{ for some } 1 \in \mathbb{Z}$$

$$\Rightarrow a/a$$

Given that

$$a/b \text{ and } b/a$$

$$b = ak_1 \rightarrow (1) \text{ for some } k_1 \in \mathbb{Z}$$

$$b = ak_2 \rightarrow (2) \text{ for some } k_2 \in \mathbb{Z}$$

From eq (1)

$$b = ak_1$$

$$b = (bk_2)k_1 \text{ by using (2)}$$

$$b = b(k_2 k_1)$$

$$b - b(k_2 k_1) = 0$$

$$b(1 - k_2 k_1) = 0$$

$$1 - k_2 k_1 = 0 \quad ; \quad b \neq 0$$

$$k_2 k_1 = 1$$

$$k_1 = k_2 = \pm 1$$

put $k = \pm 1$ in (1)

$$a = b(\pm 1)$$

$$a = \pm b$$

(v) Given that

$$a|b \quad \text{and} \quad b|c$$

$$b = ak_1 \rightarrow (1) \quad \text{for some } k_1 \in \mathbb{Z}$$

$$c = bk_2 \rightarrow (2) \quad \text{for some } k_2 \in \mathbb{Z}$$

from eq (2)

$$c = bk_2$$

$$c = (ak_1)k_2 \quad \text{by using (1)}$$

$$c = a(k_1 k_2)$$

$$c = ak_3$$

$$\because k_1, k_2 \in \mathbb{Z}$$

$$\Rightarrow a|c$$

$$k_1 k_2 = k_3 \in \mathbb{Z}$$

(vi) Given that

$$a|b$$

$$\Rightarrow b = ak \quad \text{for some } k \in \mathbb{Z}$$

$$bc = (ak)c$$

$$bc = a(kc)$$

$$kc = k_1 \in Z$$

$$bc = ak_1$$

$$~~kc =~~$$

$$\Rightarrow a|bc$$

(vii) Given that

$$a|b$$

$$\Rightarrow b = ak \text{ for some } k \in Z$$

$$\Rightarrow bc = (ak)c$$

$$\Rightarrow bc = (ac)k$$

$$\Rightarrow ac|bc$$

(viii) Given that

$$a|b \text{ and } a|c$$

$$b = ak_1 \rightarrow (1)$$

$$c = ak_2 \rightarrow (2)$$

for some $k_1 \in Z$

for some $k_2 \in Z$

Multiplying (1) by a

Multiplying (2) by a

$$ba = a^2 k_1$$

$$ca = a^2 k_2$$

$$ba = a(k_1 a)$$

$$ca = a(k_2 a)$$

$$\therefore x, y \in Z$$

$$ba = ak_3$$

$$ca = ak_4$$

Adding both eq's...

$$ba + ca = ak_3 + ak_4$$

$$ba + ca = a(k_3 + k_4)$$

$$bx + cy = aks$$

$$a \mid bx + cy$$

$$\therefore ka + ka = ksa \in \mathbb{Z}$$

Exercise

If $a \mid b_1 + b_2$ and $a \mid b_1$, then $a \mid b_2$.

Given

$$a \mid b_1 + b_2 \quad \text{and} \quad a \mid b_1$$

$$\therefore a \mid b \quad \Rightarrow \quad b = ac \quad \text{for some } c \in \mathbb{Z}$$

$$b_1 + b_2 = ak_1 \quad \rightarrow (1)$$

$$\text{for some } k \in \mathbb{Z}$$

$$\Rightarrow b_1 = ak_2 \quad \rightarrow (2)$$

$$\text{for some } k_2 \in \mathbb{Z}$$

From (2)

$$b_1 + b_2 = ak_1$$

$$ak_2 + b_2 = ak_1 \quad (\text{using (2)})$$

$$b_2 = ak_1 - ak_2$$

$$b_2 = a(k_1 - k_2)$$

$$b_2 = ak_3$$

$$\therefore k_1, k_2 \in \mathbb{Z}$$

$$\Rightarrow k_1 - k_2 = k_3 \in \mathbb{Z} \quad \Rightarrow a \mid b_2$$

Alternative Solution:

Given

$$a|b_1+b_2 \quad \text{and} \quad a|b_1$$

$$\Rightarrow a|(b_1+b_2)x + b_1y$$

In particular

$$x=1, y=-1$$

$$\Rightarrow a|b_1+b_2(-1) + b_1(1)$$

$$= b_1+b_2-b_1$$

$$\Rightarrow a|b_2$$

Example (1):

Show that $a-b|a^n-b^n$,

where $a, b, n \in \mathbb{Z}$ with $a \neq b$

$n \geq 0$.

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let

$S(n)$:

$$S(n): a-b|a^n-b^n$$

i) $S(0)$ True

Step - I:

(ii) $S(k)$

$$n=0$$

$$\Rightarrow S(k) \Rightarrow S(k+1)$$

$$a^n-b^n = a^0-b^0 = 1-1=0$$

$$\Rightarrow a-b|0$$

$$\Rightarrow S(0) \text{ is True}$$

Step - II:

let

$$S(k) = a-b \mid a^k - b^k \rightarrow (2)$$

$$S(k+1) = a-b \mid a^{k+1} - b^{k+1} \rightarrow (2)$$

From eq (1)

$$a^k - b^k = (a-b)m \rightarrow (3)$$

for some $m \in \mathbb{Z}$

Consider

$$\begin{aligned} a^{k+1} - b^{k+1} &= a a^k - b b^k \\ &= a a^k - a b^k + a b^k - b b^k \\ &= a(a^k - b^k) + b^k(a - b) \end{aligned}$$

By using eq (3)

$$\begin{aligned} a^{k+1} - b^{k+1} &= a(a-b)m + b^k(a-b) \\ &= (a-b)(am + b^k) \end{aligned}$$

Say $am + b^k = t \in \mathbb{Z}$

$$a^{k+1} - b^{k+1} = (a-b)t$$

$$\Rightarrow a-b \mid a^{k+1} - b^{k+1}$$

$\Rightarrow S(k+1)$ is true

So; $a-b \mid a^n - b^n$ for all $n \geq 0$.

(2) Show that $a+b \mid a^n + b^n$, where $a, b, n \in \mathbb{Z}$ with $a \neq -b$ and n is a positive odd integer.

Let

$$S(n): a+b \mid a^n + b^n$$

Step-1:

$$n=1$$

$$a^1 + b^1 = a+b \quad (S(1)) \text{ is true.}$$

Step-2:

let

$$S(k) = a+b \mid a^k + b^k \rightarrow (1)$$

$$S(k+2) = a+b \mid a^{k+2} + b^{k+2} \rightarrow (2)$$

from (1)

$$a^k + b^k = (a+b)^m \rightarrow (3)$$

for some $m \in \mathbb{Z}$

Consider

$$a^{k+2} + b^{k+2} = a^2 a^k + b^2 b^k$$

$$= a^2 a^k + a^2 b^k + a b^k + b^2 b^k$$

$$= a^2(a^k + b^k) + b^k(a^2 + b^2)$$

$$= a^2(a+b)^m + b^k(a^2 + b^2)$$

$$= a^2 m(a+b) + b^k(a^2 + b^2)$$

$$a^{k+2} + b^{k+2} = (a+b) [a^2 m + b^k(a-b)]$$

$$\therefore a, b, m \in \mathbb{Z}$$

$$\therefore a^2 m + b^k(a-b) \Rightarrow t \in \mathbb{Z}$$

$$a^{k+2} + b^{k+2} = (a+b)t$$

$$\Rightarrow (a+b) \mid a^{k+2} + b^{k+2}$$

$\Rightarrow S(k+2)$ is true.

Question:

Show that for all
n is positive integer $24 \mid 2 \cdot 7^n + 3 \cdot 5^n - 5$.

Step-1: $n=1$

$$2 \cdot 7^n + 3 \cdot 5^n - 5 = 2 \cdot 7^1 + 3 \cdot 5^1 - 5$$

$$= 14 + 15 - 5$$

$$= 24$$

$$\Rightarrow 24 \mid 24 = 2 \cdot 7^n + 3 \cdot 5^n - 5$$

$S(1)$ is true.

Step-2: Let

$$S(k) : 24 \mid 2 \cdot 7^k + 3 \cdot 5^k - 5 \rightarrow (1)$$

$$S(k+1) : 24 \mid 2 \cdot 7^{k+1} + 3 \cdot 5^{k+1} - 5$$

From eq (1)

$$2 \cdot 7^k + 3 \cdot 5^k - 5 = 24t \rightarrow (2)$$

for some $t \in \mathbb{Z}$

Consider

$$2 \cdot 7^{k+1} + 3 \cdot 5^{k+1} - 5 = 2 \cdot 7^k \cdot 7 + 3 \cdot 5^k \cdot 5 - 5$$

$$= 2 \cdot 7^k (6+1) + 3 \cdot 5^k (4+1) - 5$$

$$= 2 \cdot 7^k + 3 \cdot 5^k - 5 + 2 \cdot 7^k \cdot 6 + 3 \cdot 5^k \cdot 4$$

By using eq (2);

$$= 24t + 12 \cdot 7^k + 12 \cdot 5^k$$

$$= 24t + 12(7^k + 5^k)$$

$$= 24t + 24 \left(\frac{7^k + 5^k}{2} \right)$$

$$= 24 \left(t + \frac{7^k + 5^k}{2} \right)$$

As $\frac{7^k + 5^k}{2} \in \mathbb{Z}$

$$\therefore \frac{t + \frac{7^k + 5^k}{2}}{2} = m \in \mathbb{Z}$$

$$2 \cdot 7^{k+1} + 3 \cdot 5^{k+1} - 5 = 24m$$

$$\Rightarrow 24 \mid 2 \cdot 7^{k+1} + 3 \cdot 5^{k+1} - 5$$

$S(k+1)$ is true.

Well Ordering principle:

Every non-empty set of non-negative integers contains a least element.

i.e: there is some integer "a" in "S" such that $a \leq b$ for all $b \in S$.

Theorem:

Division algorithm OR euclid's theorem.

let $a, b \in \mathbb{Z}$ with $a > 0$, then
there exist unique
integers " q, r " such
that

$$b = qa + r \text{ where } 0 \leq r < a$$

proof :

let $A = \{b - ax \mid x \in \mathbb{Z} \text{ so that } b - ax \geq 0\}$

A is non-empty

let $x = -|b| \in \mathbb{Z}$

$$-ax = a|b| \geq |b| \geq -b$$

$$\Rightarrow -ax \geq -b$$

$$b - ax \geq 0 \text{ where } x = -|b|$$

$\Rightarrow A \neq \emptyset$ OR A is non-empty.

Being the non-empty sub-set
of the set " $\mathbb{W}$ ", A has
a minimal or least element

(say r)

Now $r \in A$

$\Rightarrow r = b - aq$ for some
 $q \in \mathbb{Z}$.

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$$aq + r = b$$

$$b = aq + r$$

Since $r \in A$, then $r \geq 0$

contrary Suppose $r < a$

$$\Rightarrow r \geq a$$

$$\Rightarrow r - a \geq 0$$

$$\Rightarrow r_1 = r - a \geq 0$$

Note that " $r_1 < r$ "

$$\text{again } r_1 = r - a \geq 0$$

$$r_1 = b - aq - a \geq 0$$

$$r_1 = b - a(q+1) \geq 0$$

" $r_1 \in A$ ", with $r_1 < r$

where ' r ' is the minimal element of ' A ', Contradiction, we must have $r \geq a$

$$\text{i.e. :- } 0 \leq r < a$$

Uniqueness:

We have found $a, r \in \mathbb{Z}$ such that

$$b = aq + r \text{ where } 0 \leq r < a$$

If there exist ($\exists$) some

$$a_1, r_1 \in \mathbb{Z}$$

Such that

$$b = aq_1 + r_1 \quad \text{where } 0 \leq r_1 < a$$

then

$$aq + r = a(q_1 + r_1)$$

$$a(q - q_1) = r_1 - r \rightarrow (1)$$

$$\text{and } 0 \leq |r - r_1| < a \rightarrow (2)$$

using (1)

$$\text{Take mod } b.o.s \quad |a(q - q_1)| = |r_1 - r|$$

$$a|q - q_1| = |r_1 - r| \quad \text{put in (2)}$$

$$0 \leq a|q - q_1| < a$$

$$\Rightarrow |q - q_1| = 0$$

$$q - q_1 = 0$$

$$q = q_1 \rightarrow \text{put in (1)}$$

$$a(q - q) = r_1 - r$$

$$0 = r_1 - r$$

$$r_1 = r$$

hence integers q and r
is unique.

Example:

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let n be any arbitrary integer, then by division algorithm, for $2 > 0$ and $n \in \mathbb{Z}$ unique integers

q and r such that

$$n = 2q + r \quad \text{where } 0 \leq r < 2$$

$$r = 0 \quad \text{or} \quad r = 1$$

$$\text{either } n = 2q + 0 \quad \text{or} \quad n = 2q + 1$$

$$n = 2q$$

Example # 2:

let n be any arbitrary integers, then by division algorithm for $3 > 0$, and

" n " there exist $\exists$ unique

integer q and r such that

$$n = 3q + r \quad \text{where } 0 \leq r < 3$$

$$r = 0, r = 1, r = 2$$

either

$$n = 3q + 0$$

$$= 3q$$

$$n = 3q + 1$$

$$n = 3q + 2$$

$$= 3q + 3 - 1$$

$$= 3(q+1) - 1$$

$$= 3q_1 - 1$$

$$= 3q_1 - 1$$

$$q+1 = q_1$$

$$q_1 = q$$

Example:

let $n, n+1, n+2$ be any
three consecutive integers

$$S(n): 6 | n(n+1)(n+2) \quad "n \in \mathbb{Z}"$$

Step-1:

$$n=1$$

$$S(1): 6 | 1(1+1)(1+2)$$

$$6 | 1(2)(3)$$

6/6
 $S(1)$ is true.

Step - 2:

$$n = k$$

$$S(k): 6 \mid k(k+1)(k+2) \rightarrow (1)$$

$$S(k+1): 6 \mid (k+1)(k+2)(k+3) \rightarrow (2)$$

From eq (1)

$$k(k+1)(k+2) = 6t \rightarrow (3) \text{ for some } t \in \mathbb{Z}.$$

Consider

$$(k+1)(k+2)(k+3) = k(k+1)(k+2) + 3(k+1)(k+2)$$

Using eq (3)

$$= 6t + 3(k+1)(k+2)$$

$$= 6t + 6 \frac{(k+1)(k+2)}{2}$$

$$= 6 \left[t + \frac{(k+1)(k+2)}{2} \right]$$

$$\text{AS } \frac{(k+1)(k+2)}{2} \in \mathbb{Z}.$$

$$\left[t + \frac{(k+1)(k+2)}{2} \right] = m \in \mathbb{Z}$$

$$(k+1)(k+2)(k+3) = 6m$$

$$6 \mid (k+1)(k+2)(k+3)$$

$S(k+1)$ is true.

Example:

If $a, b \in \mathbb{Z}$ and $a \neq 0$ then
 $\exists$ unique integers q, r such
that $b = aq + r$ where $0 \leq r < |a|$

Proof:

$$a \neq 0 \Rightarrow a > 0 \text{ or } a < 0$$

Case - I:

If $a > 0$ then by
division algorithm $\exists$ unique
integers q and r such that
 $b = aq + r$ where $0 \leq r < a$

so that $b = aq + r$ when $0 \leq r < |a|$

Case - 2:

$a < 0$ Then $-a > 0$ then

for b and $-a > 0$, by
division algorithm $\exists$ unique
integers q and r such that

$$b = -aq + r \text{ where } 0 \leq r < -a$$

$$b = a(-q) + r \text{ where } 0 \leq r < |a|$$

Since q and r are unique

integers therefore $-q$ and r are

unique integers.

Such that

$$b = a(-a) + r \text{ where } 0 \leq r < |a|.$$

Example:

Prove that every odd integer is of the form $4n-1$ or $4n+1$.

let m be any odd integer.

For m and 4 , by division algorithm } unique integers a, r such that

$$m = 4a + r \text{ where } 0 \leq r < 4$$

$$r = 0, 1, 2, 3.$$

$\Rightarrow$ either $m = 4a+0, 4a+1, 4a+2, 4a+3$.

$$m = 4a$$

If $m = 2(2a)$ that is impossible

because m is odd.

If $m = 4a+2 = 2(2a+1)$ then it

is impossible because m is odd.

If $m = 4a+1$, then $m = 4n+1$

where $n = a \in \mathbb{Z}$.

$$\text{If } m = \cancel{4a+3} \quad 4a+3$$

$$= 4a+4-1$$

$$= 4(a+1) - 1$$

$$m = 4n-1 \quad \text{where } n = a+1 \in \mathbb{Z}$$

Therefore either $m = 4n-1$ or $4n+1$.

Example:

If b and m are the positive integers and a and r are the quotient and remainder when a is divided by b , then show that a is quotient and mr is remainder when ma is divided by mb .

Since a and r are the quotient and remainder, when a divides by b , then by division algorithm;

$$a = ba + r \quad \text{where } 0 \leq r < a$$

$$ma = m(ba + mr) \rightarrow \text{where}$$

$$m \cdot 0 \leq mr < ma$$

$$ma = (mb)a + mr. \rightarrow$$

hence a is quotient

and mr is remainder where

ma is divided by mb .

Greatest Common

Divisor OR (H.C.F) (G.C.D)

If $a, b \in \mathbb{Z}$ with $a^2 + b^2 \neq 0$

(i.e. atleast one of a , and b

is non-zero) An integer $d \in \mathbb{Z}$

called g.c.d or H.C.F of

a and b (written as $d = (a, b)$).

(i) $d > 0$

(ii) $d|a$ and $d|b$

(iii) for any $c \in \mathbb{Z}$ with $c|a$ and

$c|b$ we must have $c|d$

Example:

(5, 15)

Divisor, of 5 = 1, 5

Divisor of $15 = 1, 3, 5, 15$

Common divisor = $1, 5$

Greatest common divisor = 5

Examples

(15, 45)

Divisors of $15 = 1, 3, 5, 15$

Divisors of $45 = 1, 3, 5, 9, 15, 45$

Common Divisor = $1, 3, 5, 15$

G.C.D = 15

Observations:

1) If $a|b$ then $(a, b) = a \cdot (\text{G.C.D})$

(2) This method is very tough and calculation based and time taken for very large integer.

(3) G.C.D of two integer exists, if one integer is non-zero.

Relatively prime integers:

Two integers $a, b \in \mathbb{Z}$ with $a^2 + b^2 \neq 0$ are said to be relatively prime or co-prime if

$$(a, b) = 1$$

Example: $(5, 7)$ are relatively prime.

$$(5, 7) = 1$$

divisor of $5 = 1, 5$

divisor of $7 = 1, 7$

Common divisor = 1

$$\text{G.C.D} = 1$$

Theorem:

If $a, b \in \mathbb{Z}$ with $a^2 + b^2 \neq 0$

then (a, b) is unique.

(G.C.D of a and b).

Proof:

Suppose $(a, b) = c$ and $(a, b) = d$

$$\Rightarrow c|a, c|b \quad \text{and} \quad d|a, d|b$$

Since $c = (a, b)$ and d is a common divisor of a and b ,

therefore;

$$d|c \rightarrow (1)$$

Since,

$d = (a, b)$ and c is a common divisor of a and b , therefore;

$$c/d \rightarrow (2)$$

From (1) and (2)

$$c = \pm d$$

but $c, d > 0$, we must have $c = d$.

hence, g.c.d of a and b are unique.

Example:

If $a, b \in \mathbb{Z}$ then $(a, b) = (-a, b) = (a, -b) = (-a, -b)$

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We know that the divisor of n and $-n$ are same for $n \in \mathbb{Z}$,

Therefore,

$$(a, b) = (-a, b) = (a, -b) = (-a, -b)$$

Theorem:

If $a, b \in \mathbb{Z}$ with $a^2 + b^2 \neq 0$

$(a, b) = d$ then $(\frac{a}{d}, \frac{b}{d}) = 1$.

Proof:

$$\text{let } \left(\frac{a}{d}, \frac{b}{d}\right) = c$$

$$\Rightarrow c > 0$$

$$c \mid \frac{a}{d} \quad \text{and} \quad c \mid \frac{b}{d}$$

$$\frac{a}{d} = ck_1 \quad \rightarrow \quad \frac{b}{d} = ck_2$$

for some $k_1, k_2 \in \mathbb{Z}$

$$\Rightarrow a = (cd)k_1, \quad b = (cd)k_2$$

$$\Rightarrow cd \mid a, \quad cd \mid b$$

$\Rightarrow cd$ is a common

divisor of a and b but

the g.c.d of a and b is d .

$$\Rightarrow cd \mid d$$

$$\Rightarrow c \mid 1$$

$$\Rightarrow c = \pm 1$$

$$c = 1 \quad \because c > 0$$

Euclidean Algorithm:

For any $a, b \in \mathbb{Z}$ with $a^2 + b^2 \neq 0$ the g.c.d of a and b exist.

Case-I:

If a and b are (+ve) integers. By division algorithm

$\exists q_1, r_1$ such that

$$b = aq_1 + r_1 \quad \text{where} \quad 0 \leq r_1 < a \rightarrow (1)$$

Assuming that $r_1 \neq 0$, By division

algorithm } a_2, r_2 such that

$$r_2 = a - r_1 a_2 \quad a = r_1 a_2 + r_2 \quad \text{where } 0 \leq r_2 < r_1 \rightarrow (2)$$

Assuming that $r_2 \neq 0$ by division algorithm } a_3, r_3 such that

$$r_3 = r_1 - r_2 a_3 \quad r_1 = r_2 a_3 + r_3 \quad \text{where } 0 \leq r_3 < r_2 \rightarrow (3)$$

Again assuming $r_3 \neq 0$ and proceeding the procedure, we can find a set

$$A = \{r_1, r_2, r_3, r_4, \dots, 0\} \text{ of remainders.}$$

Note that $A \subseteq \mathbb{N}$

If A is empty i.e. $A = \emptyset$ then we must have $r_1 = 0$ Then

$$b = a a_1 + 0$$

$$b = a a_1$$

$$\Rightarrow a|b$$

$$\Rightarrow (a, b) = a \quad (\text{exist})$$

If $A \neq \emptyset$, then A being the non-empty subset of $\mathbb{N}$ has a minimal element (say r_k) where $k \geq 1$.

Now;

$$A = [r_1 > r_2 > r_3 > r_4 > \dots > r_{k-1} > 0]$$

then

$$r_{k-2} = r_{k-1} q_k + r_k \quad \text{where}$$

$$0 \leq r_k < r_{k-1}.$$

Finally, by division algorithm

}

$q_{k+1} > r_{k+1}$ Such that

$$r_{k-1} = r_k q_{k+1} + r_{k+1} \quad \text{where } 0 \leq r_{k+1} < r_k$$

this is possible only if

$$r_{k+1} = 0. \quad \text{Otherwise } r_{k+1} \in A$$

$$\because r_{k+1} < r_k \quad \text{and} \quad \text{min } A = r_k$$

$$\Rightarrow r_{k-1} = r_k q_{k+1}$$

$$\text{claim } (a, b) = r_k$$

$$\Rightarrow r_k > 0 \quad (\text{done}).$$

Now; we have to Show

$$r_k | a, r_k | b$$

From last equation

$$r_k | r_{k-1}$$

From 2nd last equation

$$r_k | r_{k-2}$$

From eq (3)

$$r_k | r_1$$

From: ev (2)

$$r_k | a$$

From ev (3)

$$r_k | b$$

If for any $c \in \mathbb{Z}$
 $c | a, c | b$ then $c | r_k$?

If $c | a, c | b$

$\Rightarrow$ From ev (1)

$$\Rightarrow c | b - ar_1 = r_1$$

$$\Rightarrow c | r_1$$

From ev (2)

$$c | r_2$$

From ev (3)

$$c | r_3$$

From 2^{nd} last ev

$$c | r_k$$

hence

$$(a, b) = r_k$$

Case - 2:

If at least one of a and b is negative, then we know that for any $a, b \in \mathbb{Z}$

$$(a, b) = (-a, -b) = (a, -b) = (-a, b)$$

Case - 3:

If exactly one of a and b is zero.

Suppose $a=0$ and $b>0$, then $(a, b) = (0, b) = b$

Equations:

$$b = aq_1 + r_1 \Rightarrow r_1 = b - aq_1 \rightarrow (1)$$

$$a = r_1q_2 + r_2 \Rightarrow r_2 = a - r_1q_2 \rightarrow (2)$$

$$r_1 = r_2q_3 + r_3 \Rightarrow r_3 = r_1 - r_2q_3 \rightarrow (3)$$

$$r_2 = r_3q_4 + r_4 \Rightarrow r_4 = r_2 - r_3q_4 \rightarrow (4)$$

⋮

$$r_{k-2} = r_{k-1}q_k + r_k \Rightarrow r_k = r_{k-2} - r_{k-1}q_k$$

$$r_{k-1} = r_kq_{k+1} + 0 = r_kq_{k+1}$$

Linear Combinations

If $a, b \in \mathbb{Z}$ then for any $x, y \in \mathbb{Z}$ the expression $ax + by$ is called a linear combination of a and b .

e.g: $a = 3, b = 5$

take $x = -1, y = 3$

$$ax + by = 3(-1) + 5(3)$$

$$= -3 + 15$$

$$= 12$$

12 is a linear combination of 3, and 5.

Example:

Find $(243, 129)$

and write it as a linear combination of 243 and 129.

So,

$$243 = 129(1) + 114$$

$$129 = 114(1) + 15$$

$$114 = 15(7) + 9$$

$$15 = 9(1) + 6$$

$$9 = 6(1) + 3$$

129	243 129	1
114	129	7
15	114	1051
9	15	1
6	9	2
3	6	618

$$6 = 3(2) + 0$$

$$(243, 129) = 3$$

$$3 = 9 + 6(-1)$$

$$3 = 9 + (15 + 9(-1))(-1)$$

$$3 = 9 + 15(-1) + 9(2)$$

$$3 = 15(-1) + 9(2)$$

$$3 = 15(-1) + (114 + 15(-7))(2)$$

$$3 = 15(-1) + 114(2) + 15(-14)$$

$$3 = 114(2) + 15(-15)$$

$$3 = 114(2) + (129 + 114(-1))(-15)$$

$$3 = 114(2) + 129(-15) + 114(15)$$

$$3 = 129(-15) + 114(17)$$

$$3 = 129(-15) + (243 + 129(-1))(17)$$

$$3 = 129(-15) + 243(17) + 129(-17)$$

$$3 = 243(17) + 129(-32)$$

$$x = 17, y = -32$$

$$3 = 243x + 129y$$

Bezout's Theorem:

Statement:

If $a, b \in \mathbb{Z}$ not both are zero and $d = (a, b)$

then d is the smallest positive linear combination of a and b such that $d = ax + by$.

OR

Let a, b be two integers, at least one of which is non-zero then there exist integers x, y such that $(a, b) = ax + by$.

Proof:

let

$$A = \{ ax + by \mid x, y \in \mathbb{Z} \text{ such that } ax + by > 0 \}$$

A is non-empty

Since $a^2 + b^2 \neq 0$, we may assume that $a \neq 0$, then

"if $a > 0$ ", take $x = 1, y = 0$; so that $ax + by = a(1) + 0 = a > 0$

$\rightarrow a \in A$ so; $A \neq \emptyset$

If $a < 0$

take $x = -1, y = 0$

$$ax + by = a(-1) + 0$$

$$= -a > 0$$

$$\Rightarrow -a \in A \text{ so, } A \neq \emptyset.$$

$$\Rightarrow A \subseteq \mathbb{N}, A \neq \emptyset$$

So, by well-ordering principle, A has a least element (say c) Now;

$$c \in A$$

$$\Rightarrow \exists \text{ some } x, y \in \mathbb{Z}$$

$$\text{Such that } c = ax + by.$$

claim

$$d = c$$

As given

$$d = (a, b)$$

$$\Rightarrow d|a, d|b$$

$$\Rightarrow d|ax_0 + by_0 = c$$

$$\Rightarrow d|c \rightarrow (1)$$

By division algorithm, for

a and $c > 0$ $\exists$ $q, r \in \mathbb{Z}$

such that

$$a = cq + r \text{ where } 0 \leq r < c$$

$$\Rightarrow r = a - cq$$

$$r = a - (ax_0 + by_0)a$$

$$r = a - ax_0 - by_0$$

$$= a(1 - x_0) + b(-y_0)$$

If $r > 0$, then we must have

$r < c$. In this way, r being

the positive linear combination of a and b belongs to A .

$$\therefore \min A = c$$

Then we must have $r = 0$

$$\Rightarrow a = ca$$

$$\Rightarrow c|a$$

Similarly, $c|b$.

Now $c|a$ and $c|b$ where $d = (a, b)$

$$\Rightarrow c|d \rightarrow (2)$$

From (1) and (2)

$$c = \pm d$$

we must have

$$c = d$$

$$\therefore c, d > 0.$$

• G.C.D $\rightarrow$ Smallest linear combination of a and b .

(1) Is any linear combination of a and b is G.C.D?

No

$$a=10, b=4$$

$$\text{take } x=1, y=-1$$

$$ax+by = 10(1) + 4(-1) \\ = 10 - 4 = 6$$

$$(a, b) = (10, 4) = 6$$

Question 1):

If $(b, c) = 1$ and $a|c$ then
 $(a, b) = 1$

Proof:

As given

$$(b, c) = 1 \quad \exists \text{ some } x, y \in \mathbb{Z}$$

Such that

$$bx + cy = 1 \rightarrow (1)$$

As $a|c$

$$\Rightarrow c = ak \text{ for some } k \in \mathbb{Z}$$

Put in (1)

$$bx + (a.k)y = 1$$

$$bx + a(ky) = 1 \quad (\text{Associative law})$$

$$a(ky) + bx = 1$$

$$\Rightarrow (a, b) = 1$$

2) If $(a, b) = 1$ then $(a, b, a+b) = 1 \text{ or } 2$.

Proof = As given
 $(a, b) = 1$

$\exists$ Some $x, y \in \mathbb{Z}$ Such
 that

$$ax + by = 1 \rightarrow (1)$$

Let

$$(a, b, a+b) = d$$

$$\Rightarrow d | a-b, d | a+b$$

$$\text{Take } x=1, y=1$$

$$\Rightarrow d | (a-b)x + (a+b)y$$

$$\Rightarrow d | a-b + a+b$$

$$\Rightarrow d | 2a \rightarrow (2)$$

Now again $x=-1, y=1$

$$d | (a-b)x + (a+b)y$$

$$\Rightarrow d | (a-b)(-1) + (a+b)(1)$$

$$\Rightarrow d | -a+b + a+b$$

$$\Rightarrow d | 2b \rightarrow (3)$$

From (2) & (3)

$$\Rightarrow d | 2a(x) + 2b(y)$$

$$\Rightarrow d | 2(ax + by)$$

From eq (1)

$$(a, b) = d$$

$$d > 0$$

$$d | a, d | b$$

$$c | a, c | b$$

$$\Rightarrow c | d$$

$$\Rightarrow d|2$$

$$\Rightarrow d = 1, -1, 2, -2$$

$$\Rightarrow d = 1, d = 2$$

Since $d > 0$

(3) If $(a, b) = 1$ then $(a+b, a-b, ab) = 1$

Proof:

As given

$$(a, b) = 1$$

then $(a+b, a-b) = 1$ or 2

Case-1:

$$(a+b, a-b) = 1$$

$$(a+b, a-b, ab) = (1, ab) \quad \because 1|ab \\ = 1$$

Case-2:

$$(a+b, a-b) = 2$$

$$\text{then } (a+b, a-b, ab) = (2, ab)$$

~~we~~ we have to show $2 \nmid ab$

Contrary Suppose, if

$$2|ab$$

$$\Rightarrow 2|a \text{ or } 2|b \quad \because 2 \text{ is prime}$$

$$\text{But as } (a, b) = 1$$

$$\text{then } 2|a \text{ or } 2|b$$

but not both.

Assume $2|a$ but $2 \nmid b$

$$\Rightarrow a = 2k \text{ for some } k \in \mathbb{Z}$$

$$\text{Now } (a+b, a-b) = 2$$

$$\Rightarrow 2|a-b$$

By divisibility

$$\Rightarrow a-b = 2m \text{ for some } m \in \mathbb{Z}$$

$$2k - b = 2m$$

$$2k - 2m = b$$

$$2(k-m) = b$$

$$2k_1 = b$$

$$\therefore k-m = k_1$$

$$\Rightarrow 2|b, \text{ contradiction of } 2 \nmid b$$

$$\text{So, } 2 \nmid ab$$

$$= (2, ab)$$

$$= 1$$

(4) If $ax+by=m$ then $(a,b)|m$

Proof

Let

$$(a,b) = d$$

$$\Rightarrow d|a, d|b$$

$$\Rightarrow d|ax+by = m$$

$$\Rightarrow d|m$$

or $(a,b)|m$ Hence proved —
Theorem :

Let a and b be two integers atleast one of which is non-zero and $m > 0$ be an integer then
 $(ma, mb) = m(a, b)$

Proof :

Let $(a, b) = d$
 $\Rightarrow d|a, d|b$

$\Rightarrow md|ma, md|mb$

$\Rightarrow md$ is common divisor of ma and mb

$\Rightarrow md|(ma, mb) \rightarrow (1)$

Again

$(a, b) = d$

$\Rightarrow ax + by = d$ for some $x, y \in \mathbb{Z}$.

Multiply by m b.s.

$\Rightarrow m(ax + by) = m(d)$

$\Rightarrow (ma)x + (mb)y = md$

$(ma, mb) | md \rightarrow (2)$

From (1) & (2)

$$(ma, mb) = \pm md$$

$$\Rightarrow (ma, mb) = md \quad \because m > 0 \text{ and } d > 0$$

$$\Rightarrow (ma, mb) = m(a, b) \quad (\text{Proved})$$

Question: If $a = bv + r$, then
Show that $(a, b) = (b, r)$

Sol let Given $a = bv + r \rightarrow (1)$

$$d = (a, b) \text{ and } d_1 = (b, r)$$

$$\Rightarrow d|a, d|b$$

From (1)

$$r = a - bv \text{ (Linear Combination of } a \text{ \& } b)$$

$$\Rightarrow d|a - bv = r$$

$$\Rightarrow d|r$$

Now $d|b$ and $d|r$

$$d|(b, r) = d_1$$

$$\Rightarrow d|d_1 \rightarrow (2)$$

$$\text{As } d_1 = (b, r)$$

$$d_1|b, d_1|r$$

$$\Rightarrow d_1|bv + r = a$$

$$\Rightarrow d_1|a$$

Now, $d_1|a$, $d_1|b$

d_1 is common divisor of

a and b . So,

$$\Rightarrow d_1 | (a, b) = d$$

$$\Rightarrow d_1 | d \rightarrow (3)$$

From (2) & (3)

$$d = \pm d_1$$

$$d = d_1 \quad \because d > 0, d_1 > 0$$

$$\Rightarrow (a, b) = (b, r)$$

(2) Show that if $(c, b) = 1$ then

$$(ac, b) = (a, b)$$

(3) Show that if $b|a$, $c|a$ and

$$(b, c) = 1 \text{ then } bc|a. \text{ Give}$$

one example to show

that the result is not

true if $(b, c) \neq 1$.

Given

$$b|a \Rightarrow a = bk_1 \rightarrow (1)$$

$$c|a \Rightarrow a = ck_2 \rightarrow (2)$$

for some $k_1, k_2 \in \mathbb{Z}$.

$$\text{As } (b, c) = 1$$

$$bx + cy = 1 \rightarrow (3)$$

for some $x, y \in \mathbb{Z}$.

Multiply eq(3) by (a)

$$abx + acy = a$$

By using (1) and (2)

$$(ck_2)bx + (bk_1)cy = a$$

$$bc(k_2x + k_1y) = a$$

$$bc(k_3) = a \quad \because k_2x + k_1y = k_3 \in \mathbb{Z}$$

$$\Rightarrow bc | a \quad (\text{Proved})$$

Counter Example

$$b=5, c=10, a=20$$

$$(b, c) = (5, 10) = 5 \neq 1$$

$$b|a, c|a$$

$$5|20, 10|20$$

$$bc = (5)(10) = 50 \nmid 20 = a$$

i.e.: $bc \nmid a$

Q: For any integer show

$$(2a+1, 9a+4) = 1$$

$$\Rightarrow d|2a+1, d|9a+4$$

$$\Rightarrow d|(9a+4)$$

Show that if $(b, c) = 1$, then
 $(a, bc) = (a, b)(a, c)$.

As given

$$(b, c) = 1$$

$$\Rightarrow bx_0 + cy_0 = 1 \text{ for some } x_0, y_0 \in \mathbb{Z} \rightarrow (1)$$

$$\text{let } (a, bc) = d \text{ , } (a, b) = d_1$$

$$(a, c) = d_2$$

$$\} \text{ Some } x, y, x_1, y_1, x_2, y_2 \in \mathbb{Z}$$

$$ax + (bc)y = d \rightarrow (2)$$

$$ax_1 + by_1 = d_1 \rightarrow (3)$$

$$ax_2 + cy_2 = d_2 \rightarrow (4)$$

To show: $d = d_1 d_2$

$$\text{Consider } d_1 d_2 = (ax_1 + by_1)(ax_2 + cy_2) \text{ (using (3) \& (4))}$$

$$= a^2 x_1 x_2 + a c x_1 y_2 + b a y_1 x_2 + b c y_1 y_2$$

$$= a(a x_1 x_2 + c x_1 y_2 + b y_1 x_2) + b c (y_1 y_2)$$

$$\Rightarrow (a, bc) \mid d_1 d_2$$

$$\Rightarrow d \mid d_1 d_2 \rightarrow (5)$$

$$\text{As } d_1 = (a, b) \text{ and } d_2 = (a, c)$$

$$d_1 \mid a, d_1 \mid b \text{ and } d_2 \mid a, d_2 \mid c$$

$$\Rightarrow a = d_1 k_1, b = d_1 k_2, a = d_2 k_3, c = d_2 k_4$$

$$\text{for some } k_1, k_2, k_3, k_4 \in \mathbb{Z}$$

$$\text{Consider } d = d \cdot 1, d = (ax + dcy)(bx_0 + cy_0)$$

$$d = ax_0 + by_0 + cz_0 + bc^2y_0$$

$$d = ax_0 + by_0 + bc(bx_0 + cy_0)$$

$$= (d_2 k_3)(d_1 k_3)x_0 + (d_1 k_1)(d_2 k_4)y_0 +$$

$$(d_1 k_2)(d_2 k_4)(bx_0 + cy_0)$$

$$d = d_1 d_2 [k_2 k_3 x_0 + k_1 k_4 y_0 + k_2 k_4 (bx_0 + cy_0)]$$

$$d = d_1 d_2$$

$$\Rightarrow \frac{d_1 d_2}{d} \mid d \quad \rightarrow (6)$$

$$\Rightarrow d = \frac{d_1 d_2}{d} \cdot d$$

$$\Rightarrow d = d_1 d_2 \quad \because d > 0, d_1 > 0, d_2 > 0$$

$$(a, bc) = (a, b)(a, c)$$

Q: If $(a, b) = 1$ and $(a, c) = 1$ then $(a, bc) = 1$.

Given $(a, b) = 1$ } some $x, y \in \mathbb{Z}$

Such that $ax_1 + by_1 = 1 \rightarrow (1)$

and $(a, c) = 1$ } some $x_2, y_2 \in \mathbb{Z}$.

Such that $ax_2 + cy_2 = 1 \rightarrow (2)$

Multiplying (1) & (2)

$$(ax_1 + by_1)(ax_2 + cy_2) = 1$$

$$a^2 x_1 x_2 + ac x_1 y_2 + bay_1 x_2 + bc y_1 y_2 = 1$$

$$a(ax_1 x_2 + cx_1 y_2 + by_1 x_2) + bc y_1 y_2 = 1$$

$$a(ax_1 x_2 + cx_1 y_2 + by_1 x_2) + bc(y_1 y_2) = 1$$

$$ax_3 + bc y_3 = 1$$

Least Common Multiple

If $a, b \in \mathbb{Z}$ such that
 $a, b \neq 0$ then the L.C.M is
denoted by $[a, b]$ or $\angle a, b >$
and it is defined by
Some integer such that

- (1) $l > 0$
- (2) $a|l, b|l$
- (3) for any integer m with $a|m, b|m$ we must have $l|m$.

Theorem:

The L.C.M of any two
non-zero integers a and b is
unique.

Proof:

let $[a, b] = m_1$ and $[a, b] = m_2$

To show: $m_1 = m_2$

Assuming that m_1 is the L.C.M
of a and b and m_2 is (true)
common multiple of a and b , we have

$$[a, b] = m_1$$

$$a|m_1, b|m_1$$

$$m_1/m_2 \rightarrow (1) \quad (\text{def iii property})$$

Again assuming that m_2 is the L.C.M of a and b and m_1 is the (true) common multiple of a and b , we have;

$$[a, b] = m_2$$

$$\Rightarrow a/m_1, b/m_1 \rightarrow (2)$$

$$m_1/m_2 \rightarrow (2)$$

From (1) & (2)

$$m_1 = \pm m_2$$

$$m_1 = m_2 \quad \because m_1 > 0, m_2 > 0$$

Theorems

If $a, b \in \mathbb{Z}$ with $a, b \neq 0$

(both non-zero integer) then

$$a, b = |a, b|$$

$$\text{Let } (a, b) = d$$

solⁿ

$$\Rightarrow d|a, d|b$$

} some $x, y \in \mathbb{Z}$ such

$$\Rightarrow a = dk_1, b = dk_2$$

$$\text{That } ax + by = d$$

for some $k_1, k_2 \in \mathbb{Z}$

$$\text{To Show: } a, b = |a, b|$$

$$\Leftrightarrow [a, b]d = |a, b|$$

$$\Leftrightarrow [a, b] = \frac{|a, b|}{d}$$

$$[a, b] = m$$

$$\text{where } m = \frac{|ab|}{d}$$

∴ we have to check $m \geq 0$

$$\therefore |ab| \geq 0$$

$$\Rightarrow \frac{|ab|}{d} \geq 0$$

$$\Rightarrow m \geq 0$$

(ii) Now we have to check $a|m, b|m$.

$$\text{let } m = \frac{|ab|}{d}$$

$$m = \frac{|dk_1 b|}{d}$$

$$= \frac{|d||k_1||b|}{d} = \frac{d \cdot |k_1||b|}{d}$$

$$m = |k_1||b|$$

$$m \pm |k_1||b| \quad \because |x| = \pm x$$

$$\Rightarrow b|m$$

$$\Rightarrow m = \frac{|a||d||k_2|}{d} \quad \begin{matrix} \rightarrow m = |k_2||a| \\ m = \frac{d|a||k_2|}{d} \end{matrix}$$

$$m = \frac{d|a||k_2|}{d}$$

$$m = \pm |k_2|a$$

$$\Rightarrow a|m$$

(iii) let l be any positive Imp integers such that
 $a|l, b|l$

$$l = at_1, l = bt_2$$

for some $t_1, t_2 \in \mathbb{Z}$

Consider

$$\frac{l}{m} = \frac{l}{\frac{lab}{d}} = \frac{ld}{lab} = \frac{l(ax+by)}{\pm ab}$$

$$= \pm l \left(\frac{ax}{ab} + \frac{by}{ab} \right)$$

$$= \pm l \left(\frac{x}{b} + \frac{y}{a} \right)$$

$$= \pm l \left(\frac{x}{b} + \frac{ly}{a} \right)$$

$$= \pm \frac{bt_2x}{b} \pm \frac{at_1y}{a}$$

$$\frac{l}{m} = \pm t_2x + t_1y$$

$$l = m(\pm t_2x \pm t_1y)$$

$$\Rightarrow m|l$$

hence $[a, b] = m$

$$[a, b] = \frac{lab}{d}$$

$$[a, b]d = m(lab) \quad (\text{proved})$$

DATE: _____

DAY: _____

Example:

 $[273, 81]$

$$a, b = |ab| \quad 3$$

$$\begin{array}{r} 81 \overline{) 273} \\ -243 \quad 3 \\ \hline 30 \end{array}$$

$$273, 81 = |273 \times 81|$$

$$[273, 81] \cdot 3 = 273 \times 81$$

$$(273, 81) = \frac{273 \times 81}{3}$$

$$= 273 \times 27$$

$$= 7371 \text{ Ans.}$$

$$\begin{array}{r} 30 \overline{) 81} \\ -60 \quad 1 \\ \hline 21 \end{array}$$

$$\begin{array}{r} 21 \overline{) 30} \\ -21 \quad 9 \\ \hline 9 \end{array}$$

$$\begin{array}{r} 9 \overline{) 21} \\ -18 \quad 3 \\ \hline 3 \end{array}$$

$$\begin{array}{r} 3 \overline{) 9} \\ -9 \quad 0 \\ \hline 0 \end{array}$$

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